Status	Finished
Started	Thursday, 8 October 2026, 8:05 PM
Completed	Thursday, 8 October 2026, 8:14 PM
Duration	8 mins 32 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Searching

b. Complexity

c. Merging

d. Sorting

The correct answer is:
Complexity

Question **2**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Deleting data from a list

b. Arranging data in a particular format

c. Inserting data into a list

d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **3**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It allows for quicker identification of duplicates
- b. It has no effect on the analysis process
- c. It complicates the analysis process
- d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It compares adjacent elements and swaps them if necessary
- b. It builds a sorted array one element at a time
- c. It repeatedly finds the minimum element and moves it to the sorted part of the list
- d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **5**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Decreases the efficiency of algorithms
- b. Provides a basis for other algorithms to work efficiently
- c. Makes code execution slower
- d. Makes data harder to manage

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **6**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are split again into smaller sublists
- b. They are combined to form a single sorted list
- c. They are discarded
- d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **7**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort cannot handle duplicate elements
- b. Bubble Sort is less efficient for large lists
- c. Bubble Sort is difficult to implement
- d. Bubble Sort does not guarantee sorted order

The correct answer is: Bubble Sort is less efficient for large lists

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Bubble Sort
- b. Linear Search
- c. Binary Search
- d. Merge Sort

The correct answer is: Bubble Sort

Question **9**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Linear Search
b. Bubble Sort
c. Binary Search
d. Merge Sort

The correct answer is: Merge Sort

Question **10**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. All of the above
b. Selection
c. Duplicates
d. Searching

The correct answer is: All of the above

Question **11**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By using the bubble-up method
b. By sorting data in a single pass
c. By breaking the input into smaller parts and merging them
d. By comparing elements sequentially

The correct answer is: By breaking the input into smaller parts and merging them

Question **12**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Merge Sort
- b. Linear Search
- c. Bubble Sort
- d. Quick Sort

The correct answer is: Merge Sort

Question **13**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Comparing each element with the others
- b. Combining sorted sublists
- c. Sorting the entire list sequentially
- d. Dividing the input into smaller subproblems

The correct answer is: Dividing the input into smaller subproblems

Question **14**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,70,80,90,120,150,200,300,400,470
- b. 80,150,200,470,40,90,300,400,70,120
- c. 200,470,80,150,40,90,300,400,70,120
- d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **15**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Extraction

b. Selection

c. Distribution

d. Insertion

The correct answer is:
Insertion